Assignment 2: Sequence Labelling

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1 Introduction

This report addresses the task of sequence labelling for named entity recognition (NER) within an archaeological domain dataset [1]. Using a token classification approach based on the Huggingface framework [2], we aim to classify tokens into six entity types. The task involves preprocessing custom-annotated data in IOB format to enable fine-tuning of a transformer-based model. Evaluation focuses on precision, recall, and F1 scores at both the token and entity level, with macro- and micro-averaged metrics to comprehensively assess model performance across entity categories. The objective is to demonstrate effective sequence labelling and analyse the impact of label granularity on classification outcomes.

2 Data

The ArchaeoNER corpus [1] consists of English archaeological reports annotated in IOB format with six entity types: `ART` (Artefact), `CON` (Construction), `LOC` (Location), `MAT` (Material), `PER` (Period), and `SPE` (Species). We use fold 1 of the dataset, provided as three IOB files: `train.txt`, `val.txt`, and `test.txt`. To understand the data characteristics and anticipate modelling challenges, we computed basic statistics directly from the IOB files. These include the number of sentences, tokens, full entities, the proportion of non-entity (O) tokens, and the distribution over entity types. The results are shown in Tables 1 and 2.

Table 1: Dataset overview

Split	Sentences	Tokens	Entities	% O Tokens
Train	1,992	53,178	3,890	89.0%
Validation	850	17,546	972	91.1%
Test	864	17,537	803	92.9%

Table 2: Entity-type distribution

Split	ART	CON	LOC	MAT	PER	SPE
Train	1,000	1,064	342	150	1,213	121
Validation	252	163	149	40	337	31
Test	168	147	144	59	283	2

The distribution is severely imbalanced. Three entity types dominate the training data: `ART` (1,000), `CON` (1,064), and `PER` (1,213), which together account for 80% of all entities. In contrast, `LOC` has only 342 instances, `MAT` has 150, and `SPE` has just 121. The test set reflects the same pattern, with only 2 `SPE` entities. Moreover, 89.0% of training tokens are labelled O. Manual inspection shows that many entities are multi-token (e.g., ”`Celtis australis`”, ”`ragstone`”).

Two key challenges arise from the task and the data:

1. **Multi-token entity boundaries**: The IOB tagging scheme requires precise prediction of B- and I- labels for multi-word spans (e.g., `Celtis australis`). Errors in boundary detection are common in sequence labelling and directly affect full-entity evaluation.

2. **Severe class imbalance**: The `SPE` class has only 121 training instances (3.1%) and extremely high label sparsity (89.0% O tokens), making it difficult for the model to learn reliable patterns for rare entity types.

3 Results

We fine-tuned `bert-base-cased` on the training set for 3 epochs with a learning rate of 2×10^{-5}, batch size 8, and weight decay 0.01, following the default settings in the Hugging Face token classification tutorial. The model was evaluated on the validation set after each epoch, and the best checkpoint (highest validation F1) was selected for final testing on `test.txt`.

The evaluation reports (1) precision, recall, and F1-score for all B- and I-token labels, (2) precision, recall, and F1 for full entity spans, and (3) macro- and micro-averaged F1-scores over all entity types. Results are shown in Tables 3, 4, and 5.

Table 3: Precision, Recall, and F-score for B- and I-labels per entity type

Entity-type	**Label**	**Precision**	**Recall**	**F1-score**
ART	B-ART	0.5942	0.7321	0.6560
	I-ART	0.5319	0.3333	0.4098
CON	B-CON	0.4510	0.6259	0.5242
	I-CON	0.4000	0.1067	0.1684
LOC	B-LOC	0.7840	0.8819	0.8301
	I-LOC	0.8000	0.8378	0.8185
MAT	B-MAT	0.6389	0.3898	0.4842
	I-MAT	0.0000	0.0000	0.0000
PER	B-PER	0.7838	0.9223	0.8474
	I-PER	0.9417	0.8071	0.8692
SPE	B-SPE	0.1667	0.5000	0.2500
	I-SPE	0.0000	0.0000	0.0000

Table 4: Precision, Recall, and F1 for full entity spans

Entity-Type	**Precision**	**Recall**	**F1-score**
ART	0.5022	0.6845	0.5793
CON	0.4038	0.5714	0.4732
LOC	0.7041	0.8264	0.7604
MAT	0.5556	0.3390	0.4211
PER	0.7493	0.8869	0.8123
SPE	0.1667	0.5000	0.2500

Table 5: Macro and micro averaged F1 over all entities

Averaging Method	**F1-score**
Micro	0.6607
Macro	0.5494

4 Discussion

The results show clear differences in performance. These differences come from class imbalance and the challenge of predicting multi-token entities.

First, the difference between scores for different entity types tells us that frequent types are easier to learn. Table 4 shows that `PER` has a full-entity F1-score of 0.8123 and `LOC` has 0.7604. In contrast, `MAT` has 0.4211 and `SPE` has 0.2500. This happens because `PER` and `CON` appear over 1,000 times in training. But `MAT` has only 150 examples and `SPE` has 121. With few examples, the model cannot learn reliable patterns for rare types.

Second, the difference between the scores for B-labels, I-labels, and full entities tells us that boundary prediction is hard for some types but easy for others. Table 3 shows that `B-PER` F1 is 0.8474 and `I-PER` F1 is 0.8692. These are close, so full-entity F1 stays high at 0.8123. But `B-MAT` F1 is 0.4842 while `I-MAT` F1 is 0.0000. The model finds the start but never the continuation. This breaks the span, so full-entity F1 drops to 0.4211. The same problem affects `ART`: despite 1,000 training examples, I-ART F1 is only 0.4098 because artefacts are often long descriptive phrases using common words (e.g., "`Neolithic flint scraper`"). One wrong I-label makes the entire entity incorrect.

Third, the difference between macro- and micro-averaged F1 scores comes from class imbalance. Table 5 shows that micro F1 is 0.6607 while macro F1 is only 0.5494 with a gap of 0.1113 points. Micro-averaging weights each correctly predicted entity token by its frequency, so the high performance on frequent types like `PER` (F1 0.8123) and `LOC` (F1 0.7604) dominates the score. Macro-averaging gives equal weight to all six entity types, so the very low scores for `MAT` (0.4211) and `SPE` (0.2500) pull the average down significantly.

In summary, the model works well on common entity types but fails on rare ones and multi-token spans. The main problem is the lack of training data for `MAT` and `SPE`.

5 Reflection

Luis handled dataset statistics, label mapping, and final report polishing. Nallathambi implemented model fine-tuning, evaluation, score computation, and wrote the first draft.

6 Conclusion

We successfully converted the ArchaeoNER IOB data into a Hugging Face Dataset and fine-tuned `bert-base-cased` using the default token-classification hyperparameters. The model achieved a micro-averaged entity-level F1 score of 0.6607 on the test set. It performed well on frequent entity types such as PER (0.8123) and LOC (0.7604) but much more poorly on rare types such as MAT (0.4211) and SPE (0.2500). The key limitations are the severe class imbalance in the data and the model's difficulty in correctly predicting I-labels for multi-token entities. These boundary errors explain the clear drop from token-level to full-entity performance. Overall, a standard BERT model with default settings can recognise common archaeological entities reasonably well. However, it requires additional techniques such as class-balanced sampling or data augmentation to perform reliably on rare and multi-token entities.

References

[1] Alex Brandsen. Archaeo-ner-data-english: Annotated archaeological ner dataset. https://github.com/alexbrandsen/Archaeo-NER-data-English/tree/main/5-folds-test-train-val-split/fold1, 2023.

[2] Hugging Face. Token classification tutorial - huggingface course. https://huggingface.co/learn/llm-course/chapter7/2#the-conll-2003-dataset, 2023.